

Answer Key — AP Statistics Class 11: Random Variables & Expected Value

SOMATH — Teacher's edition. 60 solutions across 12 concepts.

1. What Is a Random Variable?

Q1. Yes. X is a numerical value determined by a random process. Possible values: 1, 2, 3, 4.

Q2. Possible values: 0 and 1. This is called an indicator random variable.

Q3. 0, 1, 2, 3, 4 — any whole number from 0 up to the number of flips.

Q4. False. A random variable is *any* numerical value determined by a random process — it does not have to be a whole number.

Why students often say "true": the first examples we meet are discrete counts (dice, coin flips, absences), and every value in those is a whole number. That's a feature of *those* examples, not a requirement.

Counter-examples — random variables that are not whole numbers:

- Height of a randomly chosen student: 152.3 cm, 152.34 cm, ... (continuous)
- Time to finish a test: 23.4 minutes (continuous)
- Temperature at 3 p.m.: 72.8°F (continuous)
- A student's GPA: 3.75 (discrete, but not a whole number)
- Your net winnings on the \$2 game from Section 6: $-\$2$ or $+\$3$ (discrete, integers here, but expected value $E(X) = -\$0.50$ — also a random variable's summary, not a whole number)

Rule of thumb: *discrete* random variables can be listed (often but not always whole numbers — think GPA, shoe size 8.5), and *continuous* random variables can take any value in an interval (almost never whole numbers). Both are valid random variables.

Q5. Not quite. A random variable must be *numerical*. To turn color into a random variable, we'd assign numbers (for example 1 = red, 0 = blue).

2. Discrete vs. Continuous Random Variables

Q1. Discrete — you count cars (0, 1, 2, ...).

Q2. Continuous — time can take any value in an interval (10.24 s, 10.241 s, ...).

Q3. Discrete — a countable whole number.

Q4. Continuous — weight is measured, not counted.

Q5. Discrete. The possible values can be listed — they don't form a continuum.

3. Probability Distribution of a Discrete RV

Q1. Every value 1–6 has probability $1/6$. Distribution: $P(X = k) = 1/6$ for $k = 1, 2, 3, 4, 5, 6$.

Q2. Outcomes: HH, HT, TH, TT. $P(X = 0) = 1/4$, $P(X = 1) = 1/2$, $P(X = 2) = 1/4$.

Q3. $P(X \geq 2) = P(X = 2) + P(X = 3) = 0.5 + 0.3 = 0.8$.

Q4. $P(X < 3) = P(X = 1) + P(X = 2) = 0.2 + 0.5 = 0.7$.

Q5. (1) Identify the random variable; (2) list all possible values; (3) find the probability of each value; (4) check that the probabilities add to 1.

4. Rules for a Valid Probability Distribution

Q1. $0.3 + 0.5 + 0.2 = 1.0$ and each is in $[0, 1]$. **Valid.**

Q2. Sum = $1.1 \neq 1$. **Not valid** — violates Rule 2.

Q3. Sum = 1.00 and each is in $[0, 1]$. **Valid** (uniform on 4 values).

Q4. **Not valid** — $P(X = 3) = -0.1$ violates Rule 1, even though the sum is 1.0 .

Q5. (1) Each probability satisfies $0 \leq P(X = x) \leq 1$. (2) The probabilities sum to 1: $\sum P(X = x) = 1$.

5. Finding Missing Probabilities

Q1. $0.1 + 0.4 + P + 0.2 = 1 \Rightarrow P = 1 - 0.7 = 0.3$.

Q2. $0.15 + P + 0.55 = 1 \Rightarrow P = 1 - 0.70 = 0.30$.

Q3. Sum of known = 0.60 . $P = 1 - 0.60 = 0.40$.

Q4. The sum of the given probabilities is already greater than 1, so the table is inconsistent. Check the given values — at least one is wrong.

Q5. Let $P(Z = 4) = p$. Then $P(Z = 3) = 2p$. Sum: $0.2 + 0.2 + 2p + p = 1 \Rightarrow 3p = 0.6 \Rightarrow p = 0.2$. So $P(Z = 3) = 0.4$, $P(Z = 4) = 0.2$.

6. Expected Value — The Mean of a Random Variable

Q1. $E(X) = 0(0.2) + 1(0.5) + 2(0.3) = 0 + 0.5 + 0.6 = 1.1$.

Q2. $E(X) = 1(0.5) + 2(0.3) + 5(0.2) = 0.5 + 0.6 + 1.0 = 2.1$.

Q3. Net values: +\$8 (win) or -\$2 (lose). $E = 8(0.1) + (-2)(0.9) = 0.8 - 1.8 = -\1.00 . Expect to lose \$1 per play.

Q4. $E(X) = 100(0.5) + 200(0.3) + 300(0.2) = 50 + 60 + 60 = 170$.

Q5. $E(\text{payout}) = 500(0.02) + 0(0.98) = \10 . The break-even premium is \$10 (before insurer expenses/profit).

7. Interpreting Expected Value

Q1. No — you can only observe 0, 1, 2, 3, or 4 heads. But over many sets of 4 flips, the average number of heads is 2.7.

Q2. Expected value is the **weighted average** of all outcomes (each weighted by probability). Most likely = the value with the highest probability. They can be different (and often are).

Q3. Not necessarily. Any single day's profit could be higher or lower. Over many days, the average will be close to \$450.

Q4. If you repeat the "flip 3 coins" experiment many times, the **average** number of heads per set of 3 flips will be about 1.5.

Q5. False. $E(X)$ can be any real number in the range of X , including non-attainable values like 3.5 on a die or 1.5 heads in 3 flips.

8. Expected Value in Games and Decisions

Q1. $E(X) = 5(0.3) + (-2)(0.7) = 1.5 - 1.4 = \0.10 . Positive — slightly favorable in the long run.

Q2. $E(X) = 10(0.15) + (-3)(0.85) = 1.5 - 2.55 = -\1.05 . Negative — unfavorable, don't play.

Q3. $E(\text{no warranty cost}) = 400(0.03) = \12 . With warranty guaranteed cost is \$20. $\$12 < \20 , so by expected value **skip the warranty**.

Q4. $E(\text{no warranty cost}) = 400(0.10) = \40 . $\$40 > \20 , so **buy the warranty**.

Q5. $E(B) = 20(0.6) + 45(0.4) = 12 + 18 = 30 \text{ min}$. Same expected time as A, but B has more variability. On expected value alone, they're tied.

9. Variance of a Random Variable

Q1. $\mu = 1.1$. Deviations squared: $(0-1.1)^2 = 1.21$, $(1-1.1)^2 = 0.01$, $(2-1.1)^2 = 0.81$. $\text{Var} = 1.21(0.2) + 0.01(0.5) + 0.81(0.3) = 0.242 + 0.005 + 0.243 = 0.49$.

Q2. 0. There is no variation from the mean.

Q3. Both means are 0. $\text{Var}(A) = 100$, $\text{Var}(B) = 4$. **Game A** is more spread out.

Q4. No. It is a sum of squared deviations weighted by non-negative probabilities, so it is always ≥ 0 .

Q5. $\mu = 2$. Deviations squared: 1 and 1. $\text{Var} = 1(0.5) + 1(0.5) = 1$.

10. Standard Deviation of a Random Variable

Q1. $\sigma_X = \sqrt{16} = 4$.

Q2. $\sigma_X = \sqrt{0.49} = 0.7$.

Q3. $\text{Var} = 2.5^2 = 6.25$.

Q4. $\sigma_X = \sqrt{0.75} \approx 0.87$ heads.

Q5. Because standard deviation is in the **same units** as the variable itself (dollars, minutes, cm), so it's directly interpretable. Variance is in squared units.

11. Transforming Random Variables

Q1. $\mu_Y = 7 + 10 = 17$. $\sigma_Y = 3$ (unchanged by shift).

Q2. $\mu_Y = 4(10) = 40$. $\sigma_Y = |4|(3) = 12$.

Q3. $\mu_Y = 3 + 2(20) = 43$. $\sigma_Y = |2|(5) = 10$.

Q4. $\mu_Y = 100 + (-1)(50) = 50$. $\sigma_Y = |-1|(8) = 8$.

Q5. Standard deviation is always **nonnegative**. The rule uses $|b|$, not b : $\sigma_Y = |-2| \cdot \sigma_X = 2 \cdot \sigma_X$, which is positive.

12. Combining Random Variables

Q1. $\mu_{X+Y} = 14$. $\mu_{X-Y} = 6$. Means always add or subtract.

Q2. $\sigma_{X+Y} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

Q3. Variances still ADD for a difference: $\sigma_{X-Y} = \sqrt{9 + 16} = 5$ (same as sum).

Q4. Each die: $\mu = 3.5, \sigma \approx 1.71$. Sum: $\mu = 7$, $\sigma = \sqrt{1.71^2 + 1.71^2} \approx 2.42$.

Q5. **No**. Variances add, standard deviations do NOT. Use $\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$.